# Switch-mode Soft Ultrafast Recovery Power Rectifier

## Plastic DPAK Package

# MSRD620CT, NRVSRD620VCT

State-of-the-art geometry features epitaxial construction with glass passivation and metal overlay contact. Ideally suited for low voltage, high frequency switching power supplies, free wheeling diode and polarity protection diodes.

### Features

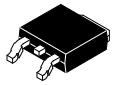
- Soft Ultrafast Recovery (35 ns typ)
- Highly Stable Oxide Passivated Junction
- Matched Dual Die Construction – May Be Paralleled for High Current Output
- Short Heat Sink Tab Manufactured – Not Sheared
- Epoxy Meets UL 94 V-0 @ 0.125 in.
- NRVSRD and SSRD8 Prefixes for Automotive and Other Applications Requiring Unique Site and Control Change Requirements; AEC-Q101 Qualified and PPAP Capable
- These Devices are Pb-Free and are RoHS Compliant*

### Mechanical Characteristics

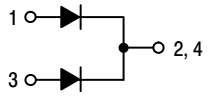
- Case: Epoxy, Molded
- Weight: 0.4 Grams (Approximately)
- Finish: All External Surfaces Corrosion Resistant and Terminal Leads are Readily Solderable
- Lead and Mounting Surface Temperature for Soldering Purposes: 260 °C Max. for 10 Seconds
- ESD Ratings:
  - Machine Model = C
  - Human Body Model = 2

* For additional information on our Pb-Free strategy and soldering details, please download the **onsemi** Soldering and Mounting Techniques Reference Manual, SOLDERRM/D.

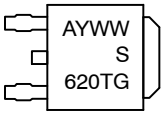
**SOFT ULTRAFAST RECTIFIER**
**6.0 AMPERES, 200 VOLTS**



**DPAK**
**CASE 369C**




### MARKING DIAGRAM




| | |
|---|---|
| A | = Assembly Location |
| Y | = Year |
| WW | = Work Week |
| G | = Pb–Free Package |

### ORDERING INFORMATION

| Device | Package | Shipping† |
|---|---|---|
| MSRD620CTT4G | DPAK (Pb-Free) | 2500 / Tape & Reel |
| NRVSRD620VCTT4G | DPAK (Pb-Free) | 2500 / Tape & Reel |

†For information on tape and reel specifications, including part orientation and tape sizes, please refer to our Tape and Reel Packaging Specifications Brochure, BRD8011/D.

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## MAXIMUM RATINGS

| Rating | Symbol | Value | Unit |
|---|---|---|---|
| Peak Repetitive Reverse Voltage<br>Working Peak Reverse Voltage<br>DC Blocking Voltage | $V_{RRM}$<br>$V_{RWM}$<br>$V_R$ | 200 | V |
| Average Rectified Forward Current<br>($T_C$ = 137 °C)<br>Per Leg<br>Per Package | $I_O$ | <br><br>3.0<br>6.0 | A |
| Peak Repetitive Forward Current<br>(Square Wave, Duty = 0.5, $T_C$ = 138 °C)<br>Per Leg | $I_{FRM}$ | <br><br>6.0 | A |
| Non−Repetitive Peak Surge Current<br>(Surge Applied at Rated Load Conditions, Halfwave, Single Phase, 60 Hz)<br>Per Package | $I_{FSM}$ | <br><br>50 | A |
| Storage / Operating Case Temperature | $T_{stg}$, $T_c$ | −55 to +175 | °C |
| Operating Junction Temperature | $T_J$ | −55 to +175 | °C |

Stresses exceeding those listed in the Maximum Ratings table may damage the device. If any of these limits are exceeded, device functionality should not be assumed, damage may occur and reliability may be affected.

## THERMAL CHARACTERISTICS

| Rating | Symbol | Value | Unit |
|---|---|---|---|
| Thermal Resistance − Junction-to-Case<br>Per Leg | $R_{\theta JC}$ | <br>9.0 | °C/W |
| Thermal Resistance − Junction-to-Ambient<br>Per Leg | $R_{\theta JA}$ | <br>80 | °C/W |

## ELECTRICAL CHARACTERISTICS

| Rating | Symbol | Value | | Unit |
|---|---|---|---|---|
| Maximum Instantaneous Forward Voltage (Note 1) (See Figure 2)<br>Per Leg | $V_F$ | **$T_J$ = 25 °C** | **$T_J$ = 150 °C** | V |
| ($I_F$ = 3.0 A)<br>($I_F$ = 6.0 A) | | 1.15<br>1.35 | 1.05<br>1.30 | |
| Maximum Instantaneous Reverse Current (See Figure 4)<br>Per Leg | $I_R$ | **$T_J$ = 25 °C** | **$T_J$ = 150 °C** | μA |
| ($V_R$ = 200 V)<br>($V_R$ = 100 V) | | 5.0<br>2.0 | 200<br>100 | |
| Maximum Reverse Recovery Time (Note 2)<br>Per Leg<br>($V_R$ = 30 V, $I_F$ = 1.0 A, di/dt = 50 A/μs)<br>($V_R$ = 30 V, $I_F$ = 3.0 A, di/dt = 50 A/μs) | $t_{rr}$ | <br><br>45<br>55 | | ns |
| Maximum Peak Reverse Recovery Current<br>Per Leg<br>($V_R$ = 30 V, $I_F$ = 1.0 A, di/dt = 50 A/μs)<br>($V_R$ = 30 V, $I_F$ = 3.0 A, di/dt = 50 A/μs) | $I_{RM}$ | <br><br>2.0<br>3.0 | | A |

1. Pulse Test: Pulse Width ≤ 250 μs, Duty Cycle ≤ 2%.
2. $t_{rr}$ measured projecting from 25% of $I_{RM}$ to ground.



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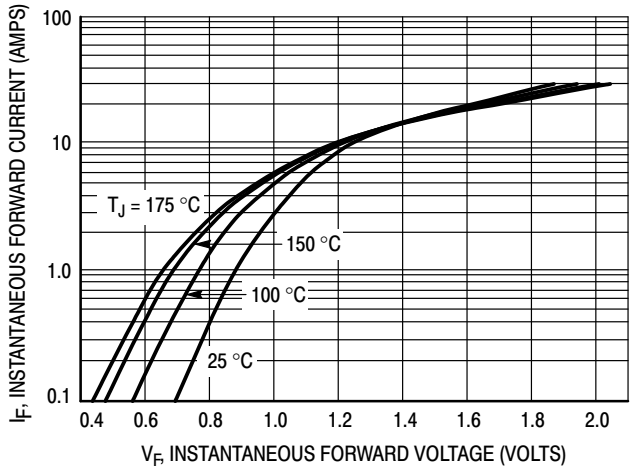



**Figure 1. Typical Forward Voltage, Per Leg**

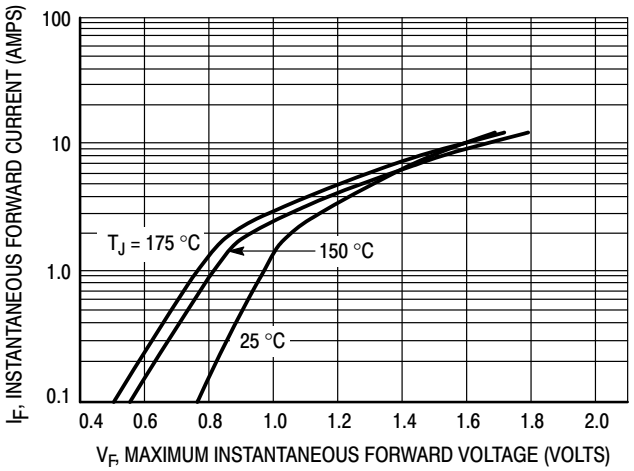



**Figure 2. Maximum Forward Voltage, Per Leg**

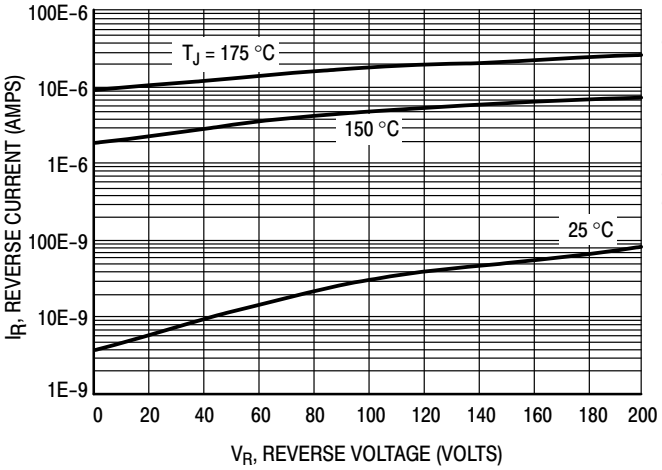



**Figure 3. Typical Reverse Current, Per Leg**

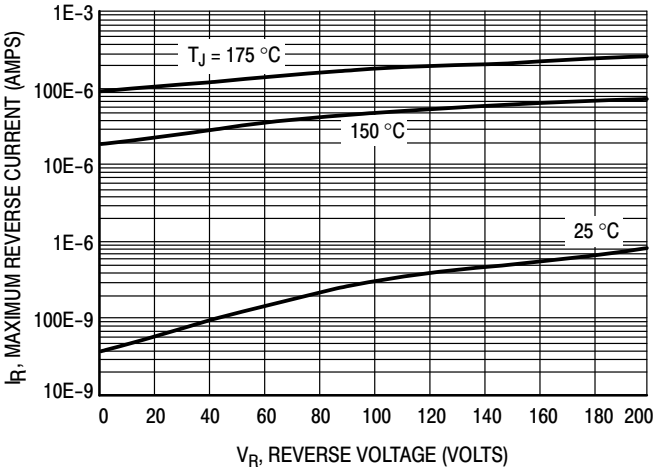



**Figure 4. Maximum Reverse Current, Per Leg**

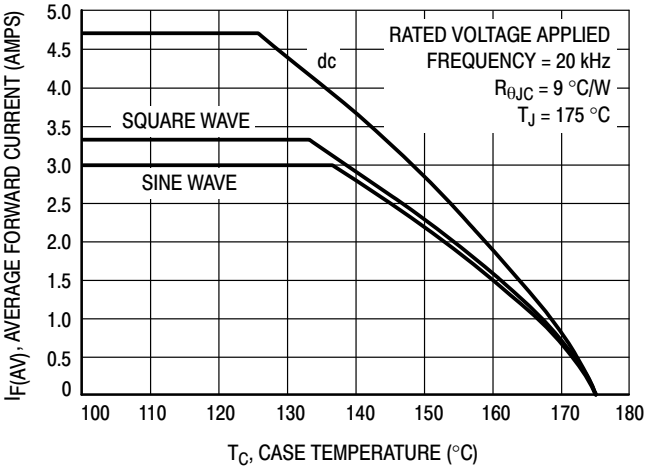



**Figure 5. Current Derating, Case (Per Leg)**

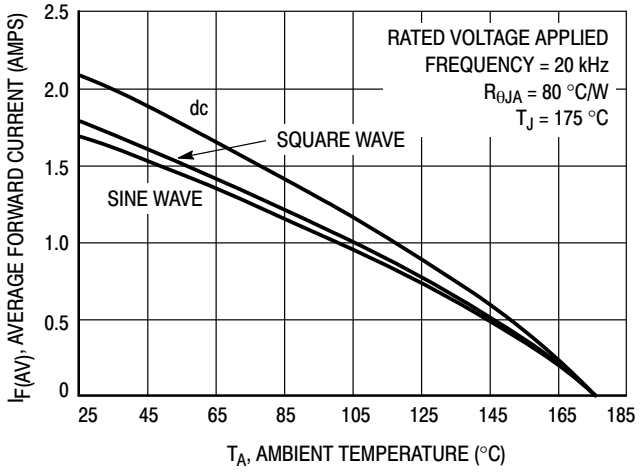



**Figure 6. Current Derating, Ambient (Per Leg)**



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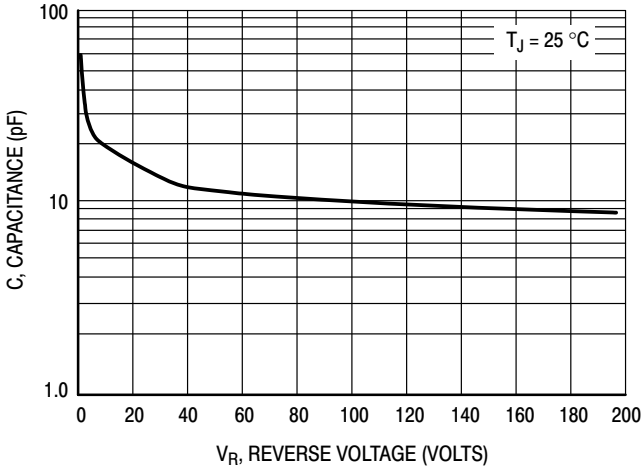



**Figure 7. Typical Capacitance (Per Leg)**

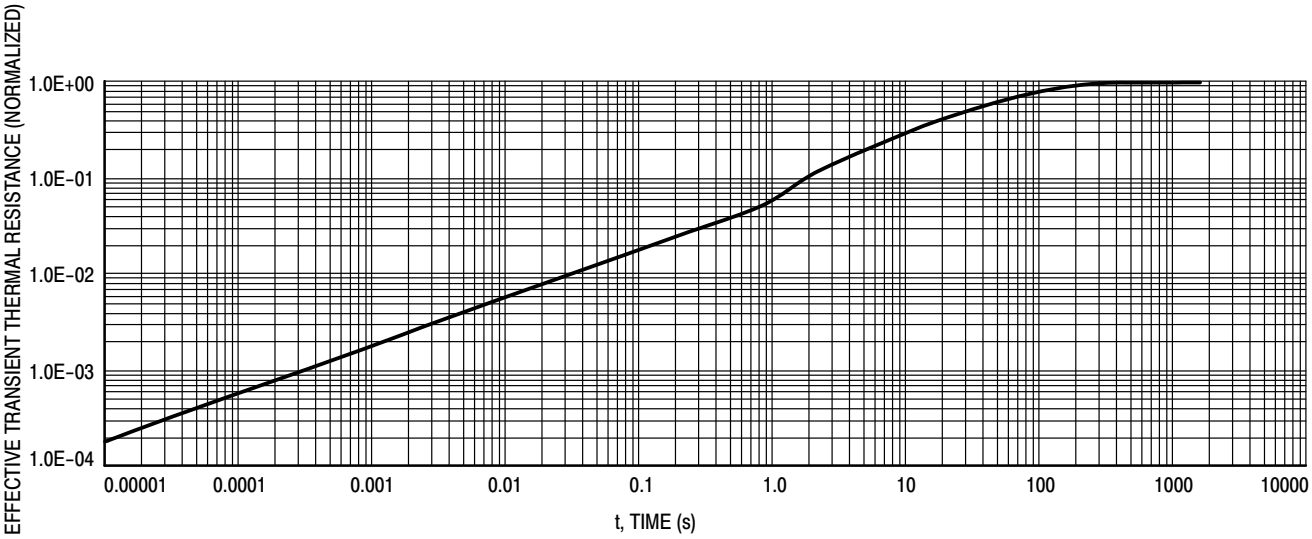



**Figure 8. Transient Thermal Response ($R_{\theta JA}$)**

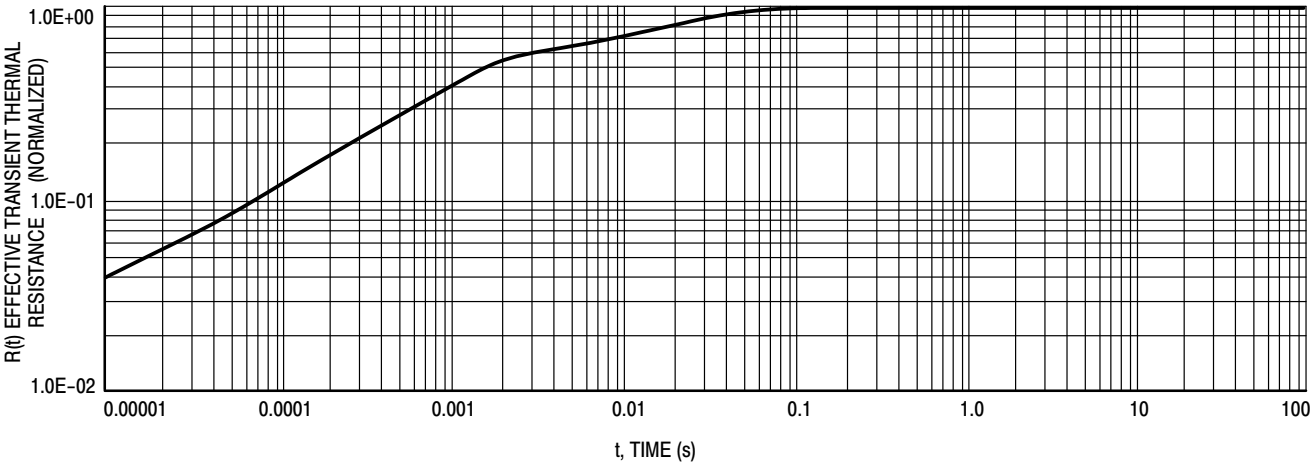



**Figure 9. Transient Thermal Response ($R_{\theta JC}$)**



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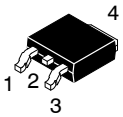
## REVISION HISTORY

| Revision | Description of Changes | Date |
|---|---|---|
| 11 | Removal of discontinued products (MSRD620CTG and SSRD8620CTT4G) from the datasheet. | 12/16/2025 |

This document has undergone updates prior to the inclusion of this revision history table. The changes tracked here only reflect updates made on the noted approval dates.



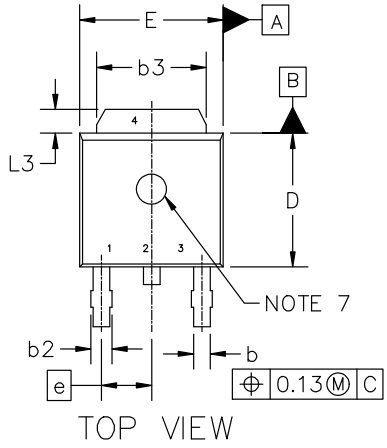
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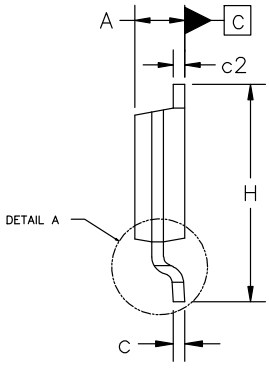

**SCALE 1:1**

**DPAK−3 6.10x6.54x2.28, 2.29P**
CASE 369C
ISSUE L

DATE 24 AUG 2026

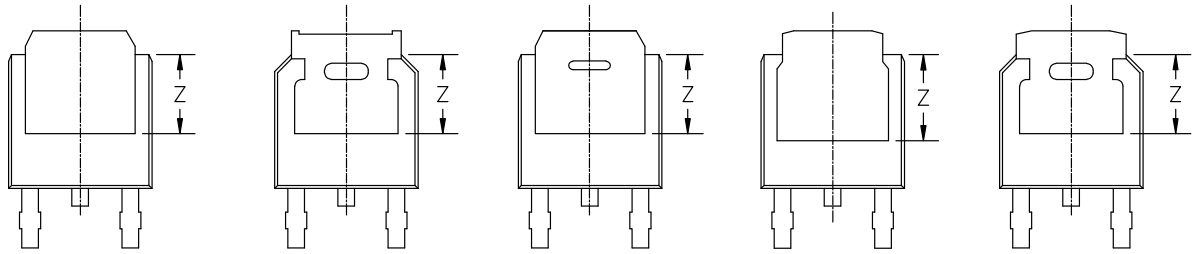



TOP VIEW




SIDE VIEW

| MILLIMETERS | | | |
|---|---|---|---|
| DIM | MIN | NOM | MAX |
| A | 2.18 | 2.28 | 2.38 |
| A1 | 0.00 | −−− | 0.13 |
| b | 0.63 | 0.76 | 0.89 |
| b2 | 0.72 | 0.93 | 1.14 |
| b3 | 4.57 | 5.02 | 5.46 |
| c | 0.46 | 0.54 | 0.61 |
| c2 | 0.46 | 0.54 | 0.61 |
| D | 5.97 | 6.10 | 6.22 |
| E | 6.35 | 6.54 | 6.73 |
| e | 2.29 BSC | | |
| H | 9.40 | 9.91 | 10.41 |
| L | 1.40 | 1.59 | 1.78 |
| L1 | 2.90 REF | | |
| L2 | 0.51 BSC | | |
| L3 | 0.89 | −−− | 1.27 |
| L4 | −−− | −−− | 1.01 |
| Z | 3.93 | −−− | −−− |

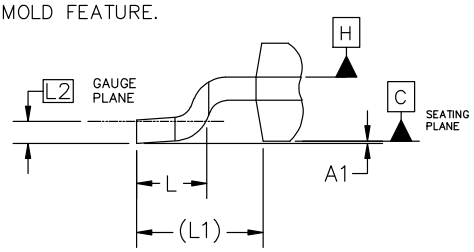



BOTTOM VIEW

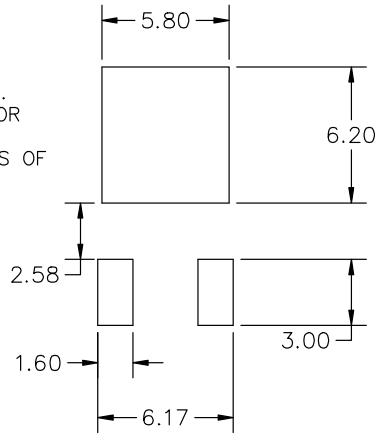
ALTERNATE CONSTRUCTIONS

NOTES:

1. DIMENSIONING AND TOLERANCING ASME Y14.5M, 2018.
2. CONTROLLING DIMENSION: MILLIMETERS.
3. THERMAL PAD CONTOUR OPTIONAL WITHIN DIMENSIONS b3, L3, AND Z.
4. DIMENSIONS D AND E DO NOT INCLUDE MOLD FLASH, PROTRUSIONS, OR BURRS.
5. DIMENSIONS D AND E ARE DETERMINED AT THE OUTERMOST EXTREMES OF THE PLASTIC BODY.
6. DATUMS A AND B ARE DETERMINED AT DATUM PLANE H.
7. OPTIONAL MOLD FEATURE.




DETAIL A
ROTATED 90° CW




RECOMMENDED MOUNTING FOOTPRINT*
*For additional information on our Pb−Free strategy and soldering details, please download the onsemi Soldering and Mounting Techniques reference manual, SOLDERRM/D.

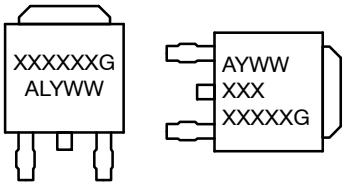
| | | |
|---|---|---|
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| **DESCRIPTION:** | **DPAK−3 6.10x6.54x2.28, 2.29P** | **PAGE 1 OF 2** |

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**DPAK−3 6.10x6.54x2.28, 2.29P**
CASE 369C
ISSUE L

DATE 24 AUG 2026

## GENERIC MARKING DIAGRAM*




**IC** **Discrete**

| | |
|---|---|
| XXXXXX | = Device Code |
| A | = Assembly Location |
| L | = Wafer Lot |
| Y | = Year |
| WW | = Work Week |
| G | = Pb−Free Package |

*This information is generic. Please refer to device data sheet for actual part marking. Pb−Free indicator, “G” or microdot “▪”, may or may not be present. Some products may not follow the Generic Marking.

STYLE 1:
PIN 1. BASE
2. COLLECTOR
3. EMITTER
4. COLLECTOR

STYLE 2:
PIN 1. GATE
2. DRAIN
3. SOURCE
4. DRAIN

STYLE 3:
PIN 1. ANODE
2. CATHODE
3. ANODE
4. CATHODE

STYLE 4:
PIN 1. CATHODE
2. ANODE
3. GATE
4. ANODE

STYLE 5:
PIN 1. GATE
2. ANODE
3. CATHODE
4. ANODE

STYLE 6:
PIN 1. MT1
2. MT2
3. GATE
4. MT2

STYLE 7:
PIN 1. GATE
2. COLLECTOR
3. EMITTER
4. COLLECTOR

STYLE 8:
PIN 1. N/C
2. CATHODE
3. ANODE
4. CATHODE

STYLE 9:
PIN 1. ANODE
2. CATHODE
3. RESISTOR ADJUST
4. CATHODE

STYLE 10:
PIN 1. CATHODE
2. ANODE
3. CATHODE
4. ANODE




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